KAIZEN MATHS

10-MINUTE PRACTICE

Algebraic substitution

Three variables

Keep track of three values and the signs between terms.

10 MINUTES / 6 QUESTIONS / AS-04

Name _____ Date _____

Show your working. Use brackets for negative values.

1 Given $a = 3, b = 3, c = 1$

Evaluate $3a - 6b + 2c$

2 Given $a = 0, b = 0, c = 0$

Evaluate $5a - 4b + 2c$

3 Given $a = 0, b = -1, c = -2$

Evaluate $4a - 4b + 4c$

4 Given $a = 3, b = -4, c = -3$

Evaluate $6a - 3b + 5c$

5 Given $a = 3, b = 0, c = -5$

Evaluate $4a - 3b + 5c$

6 Given $a = 1, b = 3, c = -3$

Evaluate $4a - 3b + 5c$

Answer sheet

Three variables | 6 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 -7

02 0

03 -4

04 15

05 -13

06 -20

Worked solutions

Three variables

01 Given $a = 3, b = 3, c = 1$

$3a - 6b + 2c$

Substitute values.

$3\times 3 - 6\times 3 + 2\times 1$

$9 - 18 + 2$

-7

02 Given $a = 0, b = 0, c = 0$

$5a - 4b + 2c$

Substitute values.

$5\times 0 - 4\times 0 + 2\times 0$

0

0

03 Given $a = 0, b = -1, c = -2$

$4a - 4b + 4c$

Substitute values.

$4\times 0 - 4\times (-1) + 4\times (-2)$

$0 + 4 - 8$

-4

04 Given $a = 3, b = -4, c = -3$

$6a - 3b + 5c$

Substitute values.

$6\times 3 - 3\times (-4) + 5\times (-3)$

$18 + 12 - 15$

15

Worked solutions

Three variables

05 Given $a = 3, b = 0, c = -5$

$4a - 3b + 5c$

Substitute values.

$4\times 3 - 3\times 0 + 5\times (-5)$

$12 - 25$

-13

06 Given $a = 1, b = 3, c = -3$

$4a - 3b + 5c$

Substitute values.

$4\times 1 - 3\times 3 + 5\times (-3)$

$4 - 9 - 15$

-20